

Sristi Bafna

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EDUCATION

- **Johns Hopkins University**
MSE Data Science; GPA: 3.6
 - **Ashoka University**
BSc.(Hons) Computer Science; GPA: 3.6
- Baltimore, USA
August 2025 - May 2027

Sonipat, India
August 2020 - May 2024

SKILLS SUMMARY

- **Languages & Frameworks:** Python, SQL, Java, React, TensorFlow, PyTorch, LangChain, and Hugging Face
- **AI/LLM Tools & Agents:** Prompt Engineering, ChatGPT (GPT-4o), Claude 3.5, Llama 3.3, Gemini, OpenAI API
- **Cloud & MLOps:** AWS (S3, EC2), Azure, Docker, and Git
- **Data & Analytics:** Pandas, Tableau, Power BI, and SQL

INDUSTRY EXPERIENCE

- **Data Analyst | Karya** Jan 2025 - June 2025
 - Built one of the first **multidimensional poverty indicators** at the district level for India, covering 700+ districts (90% across 6 pillars (social & gender inequality; infrastructure & access; education; health; climate, land & agriculture; economic status), leveraging NFHS-5, SHRUG and Climate Vulnerability Index.
 - **Reduced ad-hoc query turnaround by 80%** by **launching Metabase in 3 weeks** with 5 dashboards (payments, engagement, tasks, profiles), creating a go-to resource for team leads and automating general data requests.
 - Built **PostgreSQL reports** and wrote **automation scripts** to retrieve critical metrics (worker performance, task status, operational KPIs), streamline data-processing workflows, and ensure data consistency – cutting manual effort by 60% and error rates by 50%.
- **Machine Learning Engineer | Selkea.ai** Oct 2024 - Jan 2025
 - Built an **end-to-end ingestion & embedding pipeline** in Python to process **5,000+ RBI documents** – scraping, text extraction (HTML/PDF), multi-granular embeddings (doc, paragraph, sentence, word) with **Sentence Transformers, BERT, GPT-4**, persisted in **ChromaDB** and **AWS S3** – powering **RAG workflows** with 35% higher retrieval accuracy.
 - **Boosted regulatory traceability by 60%** by engineering a **relationship graph tool (BeautifulSoup, GPT-4, NetworkX)** to extract and visualize **semantic links** (repeals, amendments, references) across **2,000+ RBI documents**.
 - Engineered a **real-time compliance alert pipeline (Python, Selenium, ChromaDB, Flask-SocketIO)** scraping **2,000+ RBI pages** daily and streaming updates via **WebSockets/REST API** – ensuring immediate visibility of new regulations, **automatic knowledge base updates**, and dynamic graphing of relationships (e.g., repeals, amendments).
- **Lead Research Assistant | Mozilla Foundation** June 2024 - Jan 2025
 - Directed a \$25K Mozilla x USAID Responsible Computing program, mentoring **50+ student teams in large-scale applied research** (gig economy, digital infrastructure) and delivered **security workshops** on Aadhaar, UPI, DigiYatra, and EVMs using **STRIDE/DREAD frameworks**.

RESEARCH EXPERIENCE

- **Graduate Research Assistant | Computational Social Science Lab** Dec 2025 - Present
 - Built an end-to-end LLM benchmarking pipeline across AWS Bedrock and OpenAI APIs, evaluating 6 frontier models (GPT, Llama, Sonnet, Haiku, DeepSeek & Qwen) as synthetic survey respondents using automated prompt engineering, response parsing, and statistical evaluation metrics.
 - Designed multi-model evaluation framework comparing aggregate and individual-level prediction accuracy, establishing baselines and analyzing model behavior across question types on a real-world social science dataset.
- **Graduate Research Assistant | Group for Language and INTelligence** Dec 2025 - Present
 - Built probabilistic language models over structured ASL corpora to analyze information flow, predictability, and multichannel encoding in real-world narratives.
 - Translated linguistic theory into scalable computational pipelines for **modeling sequence prediction, feature extraction, and information-theoretic evaluation**.
- **Research Assistant | Mphasis Labs for Computational Thinking** April 2022 - Dec 2022
 - Analyzed tumor micro-environments in biopsy samples to support diagnosis of Triple Negative Breast Cancer (TNBC) using data from Max Healthcare Hospitals.
 - Applied supervised, unsupervised, and semi-supervised learning (ViTs, Inception-v4, ResNet-50, BIRCH, AHC), incorporating mutual-information feature selection, KL-divergence losses, and active labeling to improve diagnostic signal extraction.

PROJECTS

- **Historical Census Data Extraction (Computer Vision, Data Engineering, NLP)**: Digitized **7,000+** records from Argentina's 1893 census using DeepSeek OCR, OpenCV, and Pandas, achieving 95% time reduction and 90%+ accuracy on degraded historical documents. Tech: Python, OpenCV, Pandas, DeepSeek API, PIL, NumPy
- **Grammatical acceptability judgements in artificial neural networks (ANNs) (Natural Language Processing)**: Undergraduate thesis on assessing the ability of ANNs to understand linguistic nuances through model-specific performances, variations in input representation, and model-wise performances on specific linguistic phenomenon. Tech: Python, TensorFlow, Scikit-Learn
- **What does a single neuron compute? (Neuronal Computation)**: Final paper written as part of the PhD track of the course Introduction to Neuroscience. Discussed whether the function computed by the neuron to output a particular pulse train is arbitrary or can be partially or definitively determined. Tech: Python
- **American Sign Language Recognition (Computer Vision, Time-series processing)**: Developed a CNN and an LSTM for static and dynamic recognition of American Sign Language (ASL) as part of a course on Computer Vision. Tech: Python, TensorFlow, Scikit-Learn
- **Identifying drivers of Suicidal Ideation-Attempt Transition using an Interpretable Deep Learning Classification Model (Natural Language Processing, Deep Learning, Explainable AI)**: Drawing from the Three Step Theory of Suicide, trained BERT and RoBERTa on a Reddit dataset and used xAI algorithms (LIME and SHAP) to interpret findings. Tech: Python, TensorFlow, Scikit-Learn, HuggingFace
- **Neuroscientific Computer Vision Model (Machine Learning, Cognitive Neuroscience)**: Constructed a plausible neuroscientific approach to the training pipeline of an image classification computer vision model through curiosity and spiking neural networks as part of an Independent Study Module on Computational Cognitive Neuroscience. Tech: Python, TensorFlow, Scikit-Learn
- **NeuroCrypt: Coercion-resistant implicit memory-based authentication system (Cryptography, Cognitive Psychology)**: Devised a game with auditory and visual modalities for robust authentication systems using insights from cognitive psychology. Tech: HTML, CSS, Pixie.js, JavaScript

PUBLICATIONS

- **AAAI-22 Student Abstract: NEUROCRYPT: Coercion-Resistant Implicit Memory Authentication (Cryptography, Cognitive Psychology)**: *June 2022*

HONORS AND AWARDS

- **Hopkins Merit Scholarship** *June 2025*
Awarded \$11,500 annually in departmental tuition support. Highest in the department (~600 students).
- **Mozilla Responsible Computing Challenge** *June 2024*
Awarded \$25,000. 1 of 9 selected projects across India for embedding ethics and social context into undergrad CS curriculum.
- **Latin Honours: Cum Laude** *May 2023*
Recognized for graduating in the top 15% of the class with a cumulative GPA of 3.6+.
- **Dean's List** *May 2023*
Awarded every semester to students who maintained a 3.65+ GPA.

TEACHING EXPERIENCE

- **Artificial Intelligence and Data Analytics** Ashoka University, India
Aug 2023 - Dec 2023
Graded coursework and conducted weekly sessions for 45 students. Rated 4.7/5.
- **Cognitive Psychology** Ashoka University, India
Aug 2023 - Dec 2023
Conducted weekly sessions to assist 30 students with course content. Rated 5/5.